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ABSTRACT

We show that the generalized Boltzmann distribution is the only distribution for which the Gibbs-Shannon entropy equals the thermodynamic entropy. This result means that the thermodynamic entropy and the Gibbs-Shannon entropy are not generally equal, but rather the equality holds only in the special case where a system is in equilibrium with a reservoir.

I. INTRODUCTION

There are two well known ways to derive the Boltzmann distribution: the microcanonical derivation and the maximum entropy principle derivation. Both are found in textbooks, such as Refs. 1–5 for the microcanonical derivation and Refs. 2, 3, and 6 for the maximum entropy principle derivation. These two derivations could be naturally extended to derive the generalized Boltzmann distributions for other ensembles such as grand canonical or isothermal-isobaric ensembles. Beyond these two standard textbook derivations, the Boltzmann distribution can also be derived based on quantum dynamics.[7]

Although modern statistical thermodynamics dates back to as early as Boltzmann[8] and Gibbs,[9] new insights are still being obtained, such as the Jarzynski equality[10] and its quantum extension,[11] as well as fluctuation theorems.[12–15] There is also interest in entropy and its relationship with information, such as Refs. 16–18. In this paper, we study the basic question of the relationship between the generalized Boltzmann distribution, the thermodynamic entropy, and the Gibbs-Shannon entropy.

For context, we start by revisiting the two textbook derivations of the Boltzmann distribution.

A. The microcanonical derivation

The microcanonical derivation constructs an ensemble with fixed total energy E composed of the system of interest and a reservoir. The fundamental postulate of statistical mechanics states that the probability distribution of allowed microstates is uniform,

$$p_k = \begin{cases} \frac{1}{\Omega_E}, & E_k = E, \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where E_k is the energy of microstate k and Ω_E is the number of microstates with energy E. The interaction between the system and reservoir is assumed to be weak, in the sense that for a fixed microstate i of the system with energy E_i, the reservoir is a microcanonical ensemble with energy $E_j = E - E_i$. The assumption of weak interaction also implies that it is possible to enumerate the states of the system and reservoir independently. The probability of microstate i of the system of interest is therefore obtained by marginalizing over the allowed states j of the reservoir,

$$p_i = \sum_j \frac{1}{\Omega_E} = \frac{\Omega_{\mathrm{r}(E-E_i)}}{\Omega_E} \propto \Omega_{\mathrm{r}(E-E_i)}, \tag{2}$$

where $\Omega_{r(E-E_i)}$ is the number of states of the reservoir that satisfies $E_j = E - E_i$, which can be written as

$$\Omega_{r(E-E_i)} = \exp\left[S_{r(E-E_i)}/k_B\right] \tag{3}$$

using the definition of entropy for the microcanonical ensemble. Since the energy of the system E_s is a small fraction of the total energy, we can expand S_r around E as a power series of the system energy E_s as

$$S_{r(E-E_i)} \simeq S_{r(E)} - \left(\frac{\partial S_r}{\partial E_r}\right) E_i \simeq S_r(E) - \frac{E_i}{T}, \tag{4}$$

where $E_r = E - E_i$ is the energy of the reservoir and T is the temperature of the reservoir. This equation becomes exact in the limit of an infinite reservoir at constant temperature T.[2] Combining Eqs. (2)–(4), we get $\Omega_{r(E-E_i)} \propto \exp\left(-\frac{E_i}{k_B T}\right)$, or, equivalently, $p_i \propto \exp\left(-\frac{E_i}{k_B T}\right)$, which is the Boltzmann distribution.

It is worth mentioning that the assumption of weak interaction between the bath and the system can be relaxed by canonical typicality.[19,20]

B. The maximum entropy principle derivation

The maximum entropy principle[21,22] derives the Boltzmann distribution by maximizing the Gibbs-Shannon entropy[23] $H = -\sum_i p_i \log p_i$ under the constraints of $\langle E\rangle = \sum_i p_i E_i$ being a constant E_0 and of $\sum_i p_i = 1$. The derivation is done using the Lagrangian multiplier method which maximizes the target function

$$\mathcal{L} = -\sum_i p_i \log p_i - \beta\left(\sum_i p_i E_i - E_0\right) + \alpha\left(\sum_i p_i - 1\right), \tag{5}$$

where α and β are both Lagrangian multipliers. Zeroing the derivative of $\mathcal{L}$ with respect to p_i gives

$$0 = \frac{\delta \mathcal{L}}{\delta p_i} = -1 - \log p_i - \beta E_i + \alpha \Rightarrow p_i = C \cdot e^{-\beta E_i}. \tag{6}$$

This optimization process has been shown to be equivalent to the microcanonical derivation as shown in Subsection I A by applying the maximum entropy principle to the whole isolated system containing the system of interest and reservoir, in a two step fashion.[24]

C. Connecting the canonical ensemble with thermodynamics

After obtaining the probability distribution of the ensemble, we need to establish the connection between thermodynamic variables and ensemble quantities. For the canonical ensemble, the thermodynamic temperature T, volume V, and particle number N are naturally mapped to the T, V, N parameter of the ensemble. Since the energy of the ensemble is not a parameter but instead a random variable, the mapping of the thermodynamic internal energy U is not as obvious as it is for T, V, N.

One then introduces a new postulate that equates the thermodynamic energy U with the ensemble average of the random variable E of system energy,

$$U = \langle E\rangle = \sum_i p_i E_i = \frac{\sum_i E_i e^{-\beta E_i}}{\sum_i e^{-\beta E_i}} = -\frac{\partial \log Z}{\partial \beta}, \tag{7}$$

where $Z = \sum_i e^{-\beta E_i}$ is the partition function and $\beta = \frac{1}{k_B T}$. Since $d(\beta F) = U d\beta - \beta p dV$, i.e., βF has natural variables β and V, and $U = \frac{\partial(\beta F)}{\partial \beta}$. Comparing this relation with Eq. (7) immediately gives the equation for Helmholtz free energy $F = -k_B T \log Z$. From the definition of Helmholtz free energy $F = U - TS$, we immediately obtain that the thermodynamic entropy can then be computed as

$$S = \frac{U - F}{T} = -k_B \sum_i p_i \log p_i = k_B H, \tag{8}$$

where $H = -\sum_i p_i \log p_i$ is the Gibbs-Shannon entropy.

D. Our contribution

The above logic shows that

$$\left.\begin{matrix}\begin{pmatrix}\text{Thermodynamic}\\ \text{first law}\end{pmatrix}\\ + \\ \begin{pmatrix}\text{The Boltzmann}\\ \text{distribution}\end{pmatrix}\end{matrix}\right\} \Rightarrow \begin{pmatrix}\text{Thermodynamic entropy}\\ \text{equals}^{25}\\ \text{Gibbs-Shannon entropy}\end{pmatrix},$$

while we will now prove the opposite direction,

$$\left.\begin{matrix}\begin{pmatrix}\text{Thermodynamic}\\ \text{first law}\end{pmatrix}\\ + \\ \begin{pmatrix}\text{Thermodynamic entropy}\\ \text{equals}\\ \text{Gibbs-Shannon entropy}\end{pmatrix}\end{matrix}\right\} \Rightarrow \begin{pmatrix}\text{The Boltzmann}\\ \text{distribution}\end{pmatrix}.$$

The contribution of this paper is therefore two fold: (1) shows that the generalized Boltzmann distribution is the only distribution in which the Gibbs-Shannon entropy equals the thermodynamic entropy and (2) presents a new way of deriving the generalized Boltzmann distribution.

II. MAIN RESULT

We consider a thermodynamic system with $m + n$ generalized forces and coordinates.[26] The thermodynamic first law can be written as

$$dU = TdS + \sum_{\eta=1}^{n} X_\eta d\chi_\eta + \sum_{\eta=1}^{m} Y_\eta dy_\eta. \tag{9}$$

We use U, χ_1, ... χ_n to denote thermodynamic quantities and E, $x_1, \ldots, x_n$ to denote random variables. We also intentionally choose different letters for generalized forces (X_i and Y_i) and displacements (χ_i and y_i) to emphasize that the ensemble we are going to study is parameterized by T, generalized forces $X_1, \ldots, X_n$, and generalized displacements $y_1, \ldots, y_m$; that is, E, $x_1, \ldots x_n$ are random variables, while $T, X_1, \ldots, X_n, y_1, \ldots, y_m$ are parameters. In other words, we are studying a $(T, X_1, \ldots, X_n, y_1, \ldots, y_m)$-ensemble.

The above notation is a general framework for all thermodynamic systems. For example, for a (μ, V, T)-ensemble (one component grand canonical ensemble), we have $m = n = 1$, $y_1 = V$, $Y_1 = -p$, $\chi_1 = N$, $X_1 = \mu$.

The main result we present in this paper is the following theorem:

Theorem 1. *Consider thermodynamic systems whose first law reads like equation (9). For any $(T, X_1, \ldots, X_n, y_1, \ldots, y_m)$ ensemble that has*

1. *probability density proportional to some function of the ensemble parameters T, X_1, ..., X_n, y_1, ..., y_m and random variables E, x_1, x_2, ..., x_n,*
2. *$\langle E\rangle = U$, $\langle x_\eta\rangle = \chi_\eta$ for all $\eta = 1$, ..., n, and*
3. *the thermodynamic entropy S and the Gibbs-Shannon entropy H having a relationship $S = k_B H$,*

the probability density function of microstate $\vec{\omega}$ must have the form

$$Pr(\vec{\omega}) \propto \exp\left[\sum_{\eta=1}^{n} \frac{X_\eta x_\eta^{(\vec{\omega})}}{k_B T} - \frac{E^{(\vec{\omega})}}{k_B T}\right], \tag{10}$$

where $E^{(\vec{\omega})}, x_1^{(\vec{\omega})}, \ldots, x_n^{(\vec{\omega})}$ are the value of random variables for the state $\vec{\omega}$.

Proof. Let us begin by assuming the probability density of microstate $\vec{\omega}$ is expressed as

$$\Pr(\vec{\omega}) \propto f\left(E^{(\vec{\omega})}, x_1^{(\vec{\omega})}, \ldots, x_n^{(\vec{\omega})}; T, X_1, \ldots, y_1, \ldots\right); \tag{11}$$

the normalization constant is

$$Z = \sum_{\vec{\omega}} f_{\vec{\omega}}, \tag{12}$$

where the sum is over all microscopic states and $f_{\vec{\omega}}$ is short for $f\left(E^{(\vec{\omega})}, x_1^{(\vec{\omega})}, \ldots, x_n^{(\vec{\omega})}; T, X_1, \ldots, y_1, \ldots\right)$. In the above formula, the temperature T appears as a parameter, and microstates $\vec{\omega}$ and its macrostates $E^{(\vec{\omega})}, x_1^{(\vec{\omega})}, \ldots, x_n^{(\vec{\omega})}$ do not depend on T. Note that if $S = k_B H$, then the thermodynamic entropy

$$S = -k_B \sum_{\vec{\omega}} \frac{f_{\vec{\omega}}}{Z} \log\left(\frac{f_{\vec{\omega}}}{Z}\right) = -k_B\left\{\frac{\sum_{\vec{\omega}} f_{\vec{\omega}} \log f_{\vec{\omega}}}{Z} - \log Z\right\}. \tag{13}$$

If we perturb the temperature T by dT and keep the other parameters (X_1, ..., X_n, y_1, ..., y_m) fixed, then the change of entropy

$$dS = \left(\frac{\partial S}{\partial T}\right)_{X_1,\ldots,X_n,y_1,\ldots,y_m} dT, \tag{14}$$

where

$$\begin{aligned}\left(\frac{\partial S}{\partial T}\right)_{X_1,\ldots,X_n,y_1,\ldots,y_m} &= -k_B\left\{\sum_{\vec{\omega}} \frac{Z \cdot \frac{\partial f_{\vec{\omega}}}{\partial T} \cdot \log f_{\vec{\omega}} - \frac{\partial Z}{\partial T} \cdot f_{\vec{\omega}} \log f_{\vec{\omega}}}{Z^2}\right\} \\ &= -k_B \sum_{\vec{\omega}} \frac{\partial}{\partial T}\left(\frac{f_{\vec{\omega}}}{Z}\right) \cdot \log(f_{\vec{\omega}}).\end{aligned} \tag{15}$$

Also,

$$\langle E\rangle = \sum_{\vec{\omega}} \frac{f_{\vec{\omega}}}{Z} \cdot E^{(\vec{\omega})}, \tag{16}$$

and therefore,

$$d\langle E\rangle = \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{X_1,\ldots,X_n,y_1,\ldots,y_m} dT = \sum_{\vec{\omega}} \frac{\partial}{\partial T}\left(\frac{f_{\vec{\omega}}}{Z}\right) \cdot E^{(\vec{\omega})} \cdot dT. \tag{17}$$

The same argument applies for all $\langle x_\eta\rangle$, and we therefore have

$$d\langle x_\eta\rangle = \left(\frac{\partial\langle x_\eta\rangle}{\partial T}\right)_{X_1,\ldots,X_n,y_1,\ldots,y_m} dT = \sum_{\vec{\omega}} \frac{\partial}{\partial T}\left(\frac{f_{\vec{\omega}}}{Z}\right) \cdot x_\eta^{(\vec{\omega})} \cdot dT. \tag{18}$$

By substituting equations in condition 2 of the theorem into Eq. (9), the first law of thermodynamics can be rewritten as

$$-\frac{dS}{k_B} + \frac{d\langle E\rangle}{k_B T} - \frac{1}{k_B T}\sum_{\eta=1}^{n} X_\eta d\langle x_\eta\rangle - \frac{1}{k_B T}\sum_{\eta=1}^{m} Y_\eta dy_\eta = 0. \tag{19}$$

Substituting Eqs. (14), (15), (17), and (18) and $dy_\eta = 0$,[27] we have

$$\sum_{\vec{\omega}} \frac{\partial}{\partial T}\left(\frac{f_{\vec{\omega}}}{Z}\right) \cdot \left[\log(f_{\vec{\omega}}) + \frac{E^{(\vec{\omega})}}{k_B T} - \sum_{\eta=1}^{n} \frac{X_\eta x_\eta^{(\vec{\omega})}}{k_B T}\right] dT = 0. \tag{20}$$

Physics laws should be universal, i.e., the above equation must hold for arbitrary system, and the only way for this to happen is

$$\log(f_{\vec{\omega}}) + \frac{E^{(\vec{\omega})}}{k_B T} - \sum_{\eta=1}^{n} \frac{X_\eta x_\eta^{(\vec{\omega})}}{k_B T} = 0, \tag{21}$$

that is,

$$f_{\vec{\omega}} = \exp\left[\sum_{\eta=1}^{n} \frac{X_\eta x_\eta^{(\vec{\omega})}}{k_B T} - \frac{E^{(\vec{\omega})}}{k_B T}\right]. \tag{22}$$

We therefore obtained the generalized Boltzmann distribution. □

III. EXAMPLES

The formulation of our main result is quite general. In this section, we will show how commonly used ensembles fit into our general framework.

The single component canonical ensemble, i.e., the (N, V, T) ensemble, has $y_1 = N$, $y_2 = V$, $Y_1 = \mu$, $Y_2 = -p$. There are no χ, X, or x. The first law reads $dU = TdS - pdV + \mu dN$, and the distribution is $\Pr(\vec{\omega}) \propto \exp\left(-\frac{E^{(\vec{\omega})}}{k_B T}\right)$.

The two component grand canonical ensemble, i.e., the (μ_1, μ_2, V, T) ensemble, has $y_1 = V$, $Y_1 = -p$, $\chi_1 = N_1$, $\chi_2 = N_2$, $X_1 = \mu_1$, $X_2 = \mu_2$, $x_1^{(\vec{\omega})} = N_1^{(\vec{\omega})}$, and $x_2^{(\vec{\omega})} = N_2^{(\vec{\omega})}$. The first law reads $dU = TdS - pdV + \mu_1 dN_1 + \mu_2 dN_2$, and the distribution is $\Pr(\vec{\omega}) \propto \exp\left(\frac{\mu_1 N_1^{(\vec{\omega})} + \mu_2 N_2^{(\vec{\omega})} - E^{(\vec{\omega})}}{k_B T}\right)$.

The single component isothermal-isobaric ensemble, i.e., the (N, p, T) ensemble, has $y_1 = N$, $Y_1 = \mu$, $\chi_1 = V$, $X_1 = -p$, and $x_1^{(\vec{\omega})} = V^{(\vec{\omega})}$. The first law reads $dU = TdS - pdV + \mu dN$, and the distribution is $\Pr(\vec{\omega}) \propto \exp\left(-\frac{pV^{(\vec{\omega})}+E^{(\vec{\omega})}}{k_B T}\right)$.

IV. DISCUSSION

We showed that the ensemble for a nonisolated thermodynamic system at equilibrium described by temperature, parameters y_i, and expectation values $\langle x_i \rangle$, whose thermodynamic entropy is given by the Gibbs-Shannon formula, is necessarily described by the generalized Boltzmann's distribution. Consequently, the generalized Boltzmann distribution is the only distribution for which the Gibbs-Shannon entropy equals the thermodynamic entropy.

Unlike familiar thermodynamic quantities such as energy, volume, and pressure, the concept of thermodynamic entropy is harder to grasp. The entropy is nevertheless a well-defined state function of systems at equilibrium, stemming from the fact that $\oint đQ/T = 0$ for any closed thermodynamic path along equilibrium states. Conversely, the thermodynamic entropy is defined only for states at equilibrium.[17] So, while the question of what is the form of the entropy for states out of equilibrium is fundamentally ill-posed, it is tempting to try to extend the concept of thermodynamic entropy to stable nonequilibrium systems,[18] such as systems at steady state. Our result implies that the entropy of such systems cannot be described by the Gibbs-Shannon formula unless their distribution of states follows the generalized Boltzmann's distribution.

Boltzmann famously defined the entropy of an isolated system at energy E as $S(E) = -k_B \log p(E)$,[8] where $S(E)$ is the microcanonical entropy and $p(E) = 1/\Omega$ is the probability of occupancy of any of the degenerate microstates of the system. The identification of the thermodynamic entropy of a system in equilibrium with a reservoir with the ensemble average of the microcanonical entropy ($S = -k_B\langle \log p\rangle = -k_B \sum_i p_i \log p_i$), as in the Gibbs-Shannon entropy formula,[9] while known to be consistent with Boltzmann's distribution, appears arbitrary. For example, it is worth considering whether there could be another distribution whose Gibbs-Shannon entropy can be identified with the thermodynamic entropy. Our result shows that within the very general thermodynamic assumptions [Eqs. (9) and (11)] this is not possible.

It is also worth mentioning that, for the special case of (N, V, T) ensemble, instead of starting from the thermodynamic first law in Eq. (9), similar results could also be obtained[28] starting from the expression of heat of quantum thermodynamics,[29–32] i.e., $đQ = \sum_i E_i dp_i$. Under the condition $S = k_B H$,

$$\sum_i E_i dp_i = đQ = TdS = -k_B T \sum_i \log p_i dp_i, \tag{23}$$

which immediately gives $p_i \propto \exp\left(-\frac{E_i}{k_B T}\right)$.

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